MODULAR KOSZUL DUALITY AND THE SHADOW TOWER: ALGEBRAICITY, DEPTH CLASSIFICATION, AND THE GRAVITATIONAL COPRODUCT

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Abstract. Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular characteristic κ, cubic shadow α, and quartic shadow S_4. We prove four results. First, the weighted shadow generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ satisfies $H(t)^2 = t^4 Q(t)$ where $Q(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ is the *shadow metric*: the tower is algebraic of degree 2, and every S_r for $r \geq 5$ is determined by the three seeds (κ, α, S_4). Second, the critical discriminant $\Delta = 8\kappa S_4$ partitions the standard landscape into four depth classes: Gaussian ($r_{\max} = 2$), Lie ($r_{\max} = 3$), contact ($r_{\max} = 4$), and mixed ($r_{\max} = \infty$). Third, the genus-2 free energy is $F_2 = 7\kappa/5760$ for single-generator algebras; the first multi-channel computation for $\mathcal{W}_3$ exhibits a cross-channel correction $\delta F_2 = (c{+}204)/(16c)$, resolving the multi-generator universality question negatively. Fourth, the gravitational coproduct obtained by BRST reduction from $\widehat{\mathfrak{sl}}_2$ to Virasoro is strictly primitive at all degrees; the mechanism applies to all principal Drinfeld–Sokolov reductions.

Remark 0.1 (Relationship to companion papers). The c-independent identity $S_3(\mathrm{Vir}_c) = 2$ is now proved as a theorem; see [**?**]. The seven-face programme connecting the shadow tower to Gaudin models, Sklyanin brackets, and GZ26 commuting Hamiltonians is developed in [**?**].

1. Introduction

1.1. **The question.** Classical Koszul duality is a phenomenon of graded algebras over a field. A quadratic algebra A determines a dual coalgebra $A^!$; the bar construction $B(A)$ mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence of a linear resolution. The theory is the subject of Priddy [**?**], Beilinson–Ginzburg–Soergel [**?**], and Loday–Vallette [**?**].

On an algebraic curve X, the generators of the algebra become sections of a $\mathcal{D}_X$-module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral

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transform whose kernel is the logarithmic 1-form $\eta_{ij} = d\log(z_i - z_j)$ on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space. The resulting object, the *chiral bar complex* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$, is a factorization coalgebra on the Ran space $\mathrm{Ran}(X)$ [**?**, **?**, **?**].

At genus 0, the classical theory embeds via restriction to the formal disk (the embedding requires specifying a retraction $\mathbb{A}^1 \to \mathrm{pt}$ and its homotopy transfer; the E_1/E_∞ bar distinction, the Arnol'd algebra of $\mathrm{Conf}_n(\mathbb{A}^1)$, and the Fulton–MacPherson compactifications are already absent over a bare point). At genus $g \geq 1$, the topology of the curve forces further structure: the bar differential acquires curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$ and the scalar $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is an intrinsic invariant of the algebra called the *modular characteristic*. This curvature has no analogue in the classical setting; its geometric origin is the nontriviality of the Hodge bundle over $\overline{\mathcal{M}}_g$.

The present paper addresses a single question: *what invariants does the chiral bar construction produce, beyond the modular characteristic?*

1.2. **The answer: the shadow obstruction tower.** The bar complex on a curve of genus g with n marked points defines a pronilpotent dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ (the modular convolution algebra), whose differential encodes graph sums over stable curves. The boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ implies $D^2 = 0$ on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, and the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is therefore a Maurer–Cartan element: $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$.

The MC element $\Theta_{\mathcal{A}}$ encodes the full genus expansion. Its finite-order projections $\Theta^{\leq r}_{\mathcal{A}}$, obtained by truncating the degree filtration at level r, form the *shadow obstruction tower*. At degree 2, the tower produces the modular characteristic κ. At degree 3, a cubic shadow α (the structure constant of the transferred L_∞ bracket ℓ_3). At degree 4, a quartic shadow S_4 (the first genuinely nonlinear invariant, measuring the contact term in the four-point collision residue).

The obstruction to truncating $\Theta_{\mathcal{A}}$ at finite degree is measured by classes $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}^{\mathrm{sh}}_{r+1,0})$ in the shadow algebra $\mathcal{A}^{\mathrm{sh}} := H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$. When $o_{r+1} = 0$, the tower stops; when $o_{r+1} \neq 0$, the next shadow coefficient S_{r+1} is forced. The question is how far this process continues and what controls it.

1.3. **The shadow metric and algebraicity.** A single quadratic polynomial controls the entire shadow obstruction tower (Theorem **??**, parts (i)–(ii)).

Restrict to a one-dimensional primary slice L of the cyclic deformation complex, spanned by a single primary field η of definite conformal weight. On L, the MC recursion reduces to a scalar recursion for the shadow coefficients S_r, and the weighted generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ satisfies

$$H(t)^2 \;=\; t^4\, Q(t), \tag{1.1}$$

where the *shadow metric*

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2 \tag{1.2}$$

is a quadratic polynomial determined by the three seed invariants (κ, α, S_4). The generating function is algebraic of degree 2: $H(t) = t^2\sqrt{Q(t)}$, and every shadow coefficient S_r for $r \geq 5$ follows from the binomial expansion of $\sqrt{Q}$.

The proof converts the Maurer–Cartan recursion into a convolution identity for the Taylor coefficients of $F(t) = H(t)/t^2$, showing $F^2 = Q$. Orders $m = 0, 1, 2$ reproduce the definitions of κ, α, S_4; orders $m \geq 3$ are the recursion.

This is the *intrinsic quartic principle*: three numbers determine an infinite sequence of invariants. The analogue is the Weierstrass equation: two parameters (g_2, g_3) determine all Laurent coefficients of $\wp$, because $\wp$ satisfies an algebraic equation of degree 2 in $\wp'$. Here three parameters (κ, α, S_4) determine all S_r, because H satisfies an algebraic equation of degree 2.

1.4. **The four-class classification.** Write $Q(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$ for the *Gaussian decomposition*, where $\Delta := 8\kappa S_4$ is the *critical discriminant*. The Gaussian envelope $(2\kappa + 3\alpha t)^2$ is the part of Q visible to the quadratic and cubic shadows alone; Δ measures the irreducible quartic contribution.

The critical discriminant controls a dichotomy.

When $\Delta = 0$, the shadow metric is a perfect square: $Q = (2\kappa + 3\alpha t)^2$, $H(t) = t^2(2\kappa + 3\alpha t)$ is polynomial, and the tower terminates. If $\alpha = 0$, only $S_2 = \kappa$ survives (depth 2); if $\alpha \neq 0$, then S_2 and S_3 survive (depth 3).

When $\Delta \neq 0$, the shadow metric is irreducible: $\sqrt{Q}$ is irrational, every Taylor coefficient beyond degree 1 carries a factor of Δ, and $S_r \neq 0$ for all $r \geq 4$ generically. The tower is infinite.

Every chirally Koszul algebra whose primary-line shadow data has been computed falls into one of four *shadow depth classes* (Theorem **??**, part (iii)). The partition is exhaustive for the standard Lie-theoretic landscape (Table **??**).

Class	Δ	$r_{\max}$	Prototype	Shadow data
G (Gaussian)	0	2	Heisenberg	$\kappa = k$
L (Lie)	0	3	Affine Kac–Moody	κ, $\alpha \neq 0$
C (Contact)	0^*	4	$\beta\gamma$ system	$\kappa = 6\lambda^2 - 6\lambda + 1$
M (Mixed)	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$	κ, α, S_4

*Class **C**: $\Delta = 0$ on the primary line; the quartic contact invariant lives on a charged stratum.

Class **G**: the bar complex is classically formal, the tower terminates at κ, and all F_g follow from κ alone. Class **L**: a single cubic correction (the Lie structure constant) appears, after which the Jacobi identity kills the tower. Class **C**: the quartic contact invariant S_4 is nonzero but lives on a separate stratum whose self-bracket exits the complex, terminating the tower one

step later. Class **M**: the tower is infinite, F_g receives corrections at every degree, and the shadow coefficients S_r decay or grow at a rate governed by the *shadow growth rate*

$$\rho(\mathcal{A}) \;=\; \sqrt{\frac{9\alpha^2 + 16\kappa S_4}{4\kappa^2}}. \tag{1.3}$$

The growth rate equals the reciprocal of the branch-point modulus. When $\rho < 1$, the tower converges absolutely; when $\rho > 1$, it diverges, but $\sqrt{Q}$ extends analytically past the convergence radius. For Virasoro at central charge c:

$$\rho(\mathrm{Vir}_c) \;=\; \frac{1}{c}\sqrt{\frac{180c + 872}{5c + 22}}, \tag{1.4}$$

with a critical central charge $c_\star \approx 6.125$ (the unique positive root of $5c^3 + 22c^2 - 180c - 872 = 0$) separating the convergent and divergent regimes. At the self-dual point $c = 13$ (where the Koszul involution $\mathrm{Vir}_c \mapsto \mathrm{Vir}_{26-c}$ has a fixed point, so Vir_{13} is self-dual under uncurved quadratic Koszul duality): $\rho \approx 0.467$.

Classical Koszul duality over a field embeds into the genus-0, degree-2, $\Delta = 0$ stratum (the Gaussian case where the tower terminates immediately) via restriction to the formal disk; the embedding is not an equivalence, since the configuration-space geometry and the E_1/E_∞ distinction are already absent over a bare point. Everything beyond class **G** is new.

1.5. **The shadow connection and Koszul monodromy.** The shadow metric equips each primary line with a logarithmic connection

$$\nabla^{\mathrm{sh}} \;=\; d \;-\; \frac{Q'(t)}{2\,Q(t)}\,dt, \tag{1.5}$$

with regular singularities at the zeros of Q and residue $\frac{1}{2}$. The monodromy around each branch point is -1. This is the *Koszul sign*: the square root $\sqrt{Q}$ changes sign around a simple zero, and the shadow obstruction tower inherits this sign change through the algebraic relation $H = t^2\sqrt{Q}$.

Under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$, the shadow metric transforms by the Verdier involution. For the Virasoro family $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$, the discriminants satisfy

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) \;=\; \frac{6960}{(5c + 22)(152 - 5c)}, \tag{1.6}$$

a Virasoro-specific identity whose numerator is $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$. The dual singularities $c = -22/5$ and $c = 152/5$ sum to 26: they are Koszul partners. Other families (affine Kac–Moody, $\mathcal{W}_N$ with $N \geq 3$) produce a different numerator.

1.6. **Genus-2 free energies.** The shadow obstruction tower produces invariants of moduli spaces through graph sums over stable curves. The genus-g *free energy* $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ holds for single-generator (uniform-weight) chirally Koszul algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2}\, \lambda_g$ is the Faber–Pandharipande integral, with $\lambda_g = c_g(\mathbb{E})$ the top Chern class of the Hodge bundle and ψ_1 the cotangent line class.

At genus 2: $\lambda_2^{\mathrm{FP}} = 7/5760$, and $F_2(\mathcal{A}) = \kappa \cdot 7/5760$ for every single-generator chirally Koszul algebra. Verification requires an explicit sum over the seven stable graphs of genus 2 with no marked points (smooth genus-2 curve, figure-eight, banana, dumbbell, theta, lollipop, and chain), weighted by automorphism factors and vertex contributions from the shadow obstruction tower.

For the $\mathcal{W}_3$ algebra, which has two generators T (weight 2) and W (weight 3) producing two primary lines, the genus-2 computation is a multi-channel graph sum: each edge of the stable graph carries a channel label (T or W), each vertex contributes a shadow coefficient on the corresponding primary line, and the propagator $P_T = 2/c$, $P_W = 3/c$ depends on the channel. The per-channel diagonal sum yields $F_2^{\mathrm{scal}}(\mathcal{W}_3) = \kappa \cdot \lambda_2^{\mathrm{FP}} = 7c/6912$, but the full multi-channel graph sum produces an additional cross-channel correction $\delta F_2 = (c + 204)/(16c)$. The multi-generator universality problem is resolved negatively: the cross-channel correction is not cancelled by CohFT R-matrix terms, so $F_g \neq \kappa \cdot \lambda_g^{\mathrm{FP}}$ at genus ≥ 2 for multi-weight algebras. The correct decomposition is $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$.

The Maurer–Cartan equation on $\overline{\mathcal{M}}_{2,0}$ also produces a *planted-forest correction*

$$\delta_{\mathrm{pf}}^{(2,0)} \;=\; \frac{S_3(10\, S_3 - \kappa)}{48}, \tag{1.7}$$

measuring the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** (where $S_3 = 0$). For classes **C** and **M**, it is generically nonzero. For the Virasoro algebra: $\delta_{\mathrm{pf}}^{(2,0)} = -(c - 40)/48$.

1.7. **The gravitational coproduct.** The fourth result concerns BRST reduction. Drinfeld–Sokolov (DS) reduction takes affine Kac–Moody data to $\mathcal{W}$-algebra data via a BRST complex with ghosts; the homological perturbation lemma (HPL) transfers the A_∞ products and coproduct from the affine dg-shifted Yangian to a transferred Yangian on the $\mathcal{W}$-algebra sector.

We prove the result for $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$. The ghost-number obstruction (the homotopy h has ghost degree -1; the projection $p \otimes p$ kills all ghost-number-(-1) elements) forces the transferred coproduct to be *strictly primitive at all degrees*. The mechanism applies to any principal DS reduction $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$, since the ghost-number grading is universal; we state the general case as a

conjecture and prove it for $\widehat{\mathfrak{sl}}_2$:

$$\Delta_z^{\mathrm{Vir}}(T) \;=\; \tau_z(T)\otimes 1 + 1\otimes T, \qquad \Delta_{z,n}^{\mathrm{Vir}} = 0 \quad \text{for } n \geq 2.$$

The ghost-number mechanism kills all higher-degree coproduct corrections. The only nontrivial gravitational structure is the r-matrix twist $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, encoding the braiding of line operators.

The physical content: in Chern–Simons theory, gluons split (the Clebsch–Gordan decomposition makes the coproduct nontrivial). In 3d gravity, gravitons do not split: the coproduct is primitive, and all interactions reside in the twist.

1.8. **Relation to existing work.** The bar construction for chiral algebras was introduced by Beilinson and Drinfeld [**?**] and developed in the factorization algebra framework by Francis and Gaitsgory [**?**] and Costello and Gwilliam [**?**]. The modular characteristic κ appears implicitly in the work of Feigin and Fuchs on Hodge bundles of conformal blocks, and the formula $\kappa(\mathrm{Vir}_c) = c/2$ can be extracted from Beilinson and Schechtman [**?**]. The invariants of moduli spaces produced by vertex algebras via conformal blocks are the subject of Faber and Pandharipande [**?**], Mumford [**?**], and the extensive literature on cohomological field theories following Kontsevich and Manin [**?**].

The quartic shadow S_4, the critical discriminant Δ, the four-class partition, and the algebraicity theorem (Theorem **??**) appear to be new. The connection between shadow depth and L_∞ non-formality, while consonant with the general philosophy of Kontsevich [**?**], has not been established previously in the chiral setting.

The Drinfeld–Sokolov reduction and its homological algebra are classical [**?**, **?**, **?**]; the gravitational coproduct primitivity result (Theorem **??**) and its connection to the shadow obstruction tower appear to be new.

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The shadow metric, the Riccati algebraicity theorem, and the depth classification Raeez Lorgat

Remark 1.1 (Relationship to companion papers). The Virasoro spectral R-matrix $R(z) = z^{2h}\exp(-(c/4)/z^2)$ and its connection to the Riccati shadow structure are developed in the companion paper *The Virasoro spectral R-matrix* (`virasoro_r_matrix.tex`).

2. The shadow metric and the Riccati equation

2.1. **The shadow obstruction tower.** Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ and universal

Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$. The complex carries a degree filtration $F^\bullet$ with $F^r = \bigoplus_{s\geq r} \mathrm{Def}^{(s)}_{\mathrm{cyc}}$, and the *shadow obstruction tower* is the sequence of finite-order truncations

$$\Theta_{\mathcal{A}}^{\leq 2},\ \Theta_{\mathcal{A}}^{\leq 3},\ \Theta_{\mathcal{A}}^{\leq 4},\ \dots$$

where $\Theta_{\mathcal{A}}^{\leq r}$ solves the Maurer–Cartan equation in the quotient $\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}/F^{r+1}$. The degree-$r$ projection $\mathrm{Sh}_r(\mathcal{A})$ is the *r-th shadow* of $\mathcal{A}$. On a one-dimensional slice of the deformation complex, Sh_r is determined by a single scalar S_r; the MC equation projected to degree r gives the *all-degree master equation*

$$\nabla_H(\mathrm{Sh}_r) + \mathfrak{o}^{(r)} = 0, \tag{2.1}$$

where ∇_H is the linearized operator and $\mathfrak{o}^{(r)}$ is the obstruction from lower degrees.

2.2. **Primary lines and the shadow metric.**

Definition 2.1 (Primary line)**.** A *primary line* in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ is a one-dimensional subspace $L = \mathbf{k}\cdot\eta$ spanned by a single primary field η of definite conformal weight h_η. For a chiral algebra with generators $\phi_1, \dots, \phi_N$ of weights $h_1, \dots, h_N$, each generator spans a primary line; the full first-order deformation space is $\bigoplus_i L_i$.

Let L be a primary line with coordinate x, and write $\mathrm{Sh}_r|_L = S_r\, x^r$ for the shadow data restricted to L. Set

$$\kappa := S_2, \qquad \alpha := S_3.$$

The scalar κ is the modular characteristic (curvature) of $\mathcal{A}$ on L; the scalar α is the cubic shadow.

Definition 2.2 (Shadow metric)**.** Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* (shadow quadratic form) on L is the polynomial

$$Q_L(t) \ := \ a_0^2 + 2\,a_0\,a_1\,t + (a_1^2 + 2\,a_0\,a_2)\,t^2 \ = \ 4\kappa^2 + 12\kappa\alpha\,t + (9\alpha^2 + 16\kappa S_4)\,t^2 \ \in \ k(c)[t]. \tag{2.2}$$

The *critical discriminant* of L is

$$\Delta := 8\kappa S_4 = a_0\,a_2. \tag{2.3}$$

Remark 2.3. The shadow metric Q_L is a quadratic polynomial in t, determined by three OPE invariants: κ, α, S_4. The term "metric" refers to Q_L's role as the quadratic form governing the tower's geometry (branch points, monodromy, convergence radius), not to a Riemannian metric.

2.3. **The recursion on a primary line.** On a primary line L, the all-degree master equation (2.1) reduces to a recursion among the scalars S_r.

Proposition 2.4 (Single-line recursion)**.** *On a primary line L with propagator $P := 1/\kappa$, the MC equation at degree $r \geq 5$ is*

$$S_r \;=\; -\frac{P}{2r} \sum_{\substack{j+k=r+2 \\ 3\leq j\leq k}} c_{jk}\, j\, k\, S_j\, S_k, \qquad c_{jk} = \begin{cases} 1 & \text{if } j < k, \\ 1/2 & \text{if } j = k. \end{cases} \tag{2.4}$$

This recursion is well-defined: since $j \geq 3$ and $j+k = r+2$, every term on the right has $k = r+2-j \leq r-1$, so S_r is determined by $\{S_2, S_3, S_4, \ldots, S_{r-1}\}$. The degree-4 shadow S_4 is an intrinsic OPE invariant (the quartic contact term in the four-point collision residue), not determined by the recursion; it serves as initial data alongside $\kappa = S_2$ and $\alpha = S_3$.

Proof. The Lie bracket on the cyclic deformation complex, restricted to L, pairs $\mathrm{Sh}_j = S_j x^j$ and $\mathrm{Sh}_k = S_k x^k$ to produce a term of degree $j+k-2$ with coefficient $jk\, P\, S_j S_k$. The eigenvalue identity $\nabla_H(x^r) = 2r\, x^r$ separates the MC equation at degree r into the linear term $2r\, S_r$ on the left and the obstruction sum on the right, giving the stated recursion. The symmetry factor c_{jk} accounts for the factor $\frac{1}{2}$ in $\frac{1}{2}[\Theta, \Theta]$ when $j = k$. □

2.4. **The weighted shadow generating function.**

Definition 2.5 (Weighted shadow generating function)**.** The *weighted shadow generating function* on L is

$$H(t) \;:=\; \sum_{r\geq 2} r\, S_r\, t^r. \tag{2.5}$$

Setting $a_n := (n+2)\, S_{n+2}$ for $n \geq 0$, this becomes $H(t) = \sum_{n\geq 0} a_n\, t^{n+2}$, with initial values $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$.

The weighted generating function H packages the recursion (2.4) as a convolution identity. Set $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\, t^n$. The MC recursion forces every Cauchy coefficient of F^2 beyond degree 2 to vanish, so F^2 is a polynomial of degree 2.

3. Algebraicity and the depth classification

3.1. **The Riccati algebraicity theorem.**

Theorem 3.1 (Riccati algebraicity)**.** *Let L be a primary line with shadow data S_r, shadow metric Q_L (Definition 2.2), and weighted shadow generating function H (Definition 2.5). Then the all-degree master equation on L is equivalent to the algebraic relation*

$$H(t)^2 \;=\; t^4\, Q_L(t). \tag{3.1}$$

The shadow generating function is therefore algebraic of degree 2 over $k(\kappa, \alpha, S_4)(t)$:

$$H(t) \;=\; t^2 \sqrt{Q_L(t)}. \tag{3.2}$$

Proof. Set $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\, t^n$ where $a_n = (n+2)\, S_{n+2}$. The claim (3.1) is equivalent to $F(t)^2 = Q_L(t)$. We verify this by computing the Cauchy product $F^2 = \sum_{m\geq 0}\big(\sum_{i+j=m} a_i a_j\big)\, t^m$ and showing it agrees with Q_L at all orders.

Orders 0, 1, 2 *(initial data).* At $m = 0$: $a_0^2 = (2\kappa)^2 = 4\kappa^2$. At $m = 1$: $2\, a_0\, a_1 = 2 \cdot 2\kappa \cdot 3\alpha = 12\kappa\alpha$. At $m = 2$: $a_1^2 + 2\, a_0\, a_2 = 9\alpha^2 + 2 \cdot 2\kappa \cdot 4S_4 = 9\alpha^2 + 16\kappa S_4$. These are exactly the coefficients of $Q_L(t)$ in (2.2).

Orders $m \geq 3$ (the MC recursion forces vanishing). Since Q_L has degree 2 in t, the identity $F^2 = Q_L$ requires

$$\sum_{i+j=m} a_i\, a_j \;=\; 0 \qquad \text{for all } m \geq 3. \tag{3.3}$$

We show that this is exactly the MC recursion (2.4) at degree $r = m + 2$.

The sum $\sum_{i+j=m} a_i a_j$ with $a_i = (i+2)\, S_{i+2}$ and $a_j = (j+2)\, S_{j+2}$ equals $\sum_{p+q=m+4,\, p,q\geq 2} p\, q\, S_p\, S_q$, where $p = i+2$, $q = j+2$. Setting $r = m+2$, this becomes $\sum_{p+q=r+2,\, p,q\geq 2} p\, q\, S_p\, S_q$.

Isolating the terms with $p = 2$ or $q = 2$ (which contribute 2κ from $S_2 = \kappa$):

$$\begin{aligned}\sum_{p+q=r+2,\, p,q\geq 2} pq\, S_p S_q &= 2 \cdot 2r \cdot \kappa\, S_r + \sum_{\substack{p+q=r+2\\ p,q\geq 3}} pq\, S_p S_q \\ &= 4r\kappa\, S_r + 2 \sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_j S_k,\end{aligned}$$

where the factor 2 in front of the restricted sum accounts for $S_p S_q + S_q S_p$ when $p \neq q$, and c_{jk} corrects for the diagonal. Substituting the recursion (2.4):

$$2 \sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_j S_k \;=\; 2 \cdot (-2r\kappa\, S_r) \;=\; -4r\kappa\, S_r.$$

Hence $\sum_{p+q=r+2} pq\, S_p S_q = 4r\kappa S_r - 4r\kappa S_r = 0$, which is (3.3).

Conversely, the vanishing (3.3) at order $m \geq 3$ implies the recursion (2.4) at $r = m + 2$ by solving for S_r (using $\kappa \neq 0$).

The equivalence of (3.1) with the full MC recursion is therefore established at all degrees. Taking positive square roots (the branch with $F(0) = a_0 = 2\kappa > 0$) gives $H(t) = t^2 F(t) = t^2 \sqrt{Q_L(t)}$. □

Remark 3.2 (Riccati ODE reformulation). Set $U(t) := \sum_{r\geq 3} S_r\, t^r$ (the non-linear part, removing Sh_2). The algebraic relation (3.1) is equivalent to the Riccati-type first-order ODE

$$2t\, U'(t) + \frac{P}{2}\big(U'(t)\big)^2 = R_3\, t^3 + R_4\, t^4, \tag{3.4}$$

where $R_3 = 6\alpha$ and $R_4 = (9\alpha^2 + 2\Delta)/(2\kappa)$ encode the cubic and quartic inputs. The left-hand side is quadratic in U': this is the precise sense in which the shadow obstruction tower is algebraic of degree 2.

3.2. **Gaussian decomposition.**

Corollary 3.3 (Gaussian decomposition)**.** *The shadow metric decomposes as*

$$Q_L(t) \;=\; (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2, \tag{3.5}$$

where $\Delta = 8\kappa S_4$ is the critical discriminant. The first term $(2\kappa + 3\alpha t)^2$ is the Gaussian envelope*: the shadow metric of the truncated tower with $S_r = 0$ for $r \geq 4$. The second term $2\Delta\, t^2$ is the* interaction correction*, measuring the quartic OPE contribution not captured by the cubic data alone.*

Proof. $(2\kappa+3\alpha t)^2 = 4\kappa^2+12\kappa\alpha\, t+9\alpha^2\, t^2$. Adding $2\Delta\, t^2 = 16\kappa S_4\, t^2$ recovers the coefficient $9\alpha^2 + 16\kappa S_4$ of t^2 in $Q_L(t)$. □

3.3. **Intrinsic quartic principle.**

Corollary 3.4 (Intrinsic quartic principle)**.** *On a primary line L, the shadow obstruction tower has intrinsic OPE data at degrees 2, 3, 4 only. All shadow coefficients S_r for $r \geq 5$ are determined by (κ, α, S_4) alone, via the closed form*

$$S_r \;=\; \frac{1}{r}\,[t^{r-2}]\,\sqrt{Q_L(t)} \qquad (r \geq 2). \tag{3.6}$$

Proof. $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ depends on exactly (κ, α, S_4). Since $H(t) = t^2\sqrt{Q_L(t)} = \sum_{r\geq 2} r S_r\, t^r$ by Theorem 3.1, comparing coefficients gives $rS_r = [t^{r-2}]\sqrt{Q_L}$. □

3.4. **The shadow connection.**

Proposition 3.5 (Shadow connection)**.** *The shadow metric defines a logarithmic connection on $\mathcal{O}_L$ over $L \setminus \{Q_L = 0\}$:*

$$\nabla^{\mathrm{sh}} \;=\; d - \omega, \qquad \omega \;:=\; \frac{Q_L'(t)}{2\,Q_L(t)}\,dt \;=\; d\big(\log\sqrt{Q_L}\big). \tag{3.7}$$

(i) *The connection is flat: $(\nabla^{\mathrm{sh}})^2 = 0$.*
(ii) *At each simple zero t_0 of Q_L, the residue of ω is $\frac{1}{2}$.*
(iii) *The monodromy is $\exp(2\pi i \cdot \frac{1}{2}) = -1$: the standard monodromy of a square root around a simple branch point.*
(iv) *The flat section with $\Phi(0) = 1$ is $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)} = \sqrt{Q_L(t)}/(2\kappa)$, and the shadow generating function is $H(t) = 2\kappa\, t^2\, \Phi(t)$: the parallel transport of the curvature along the degree parameter, weighted by t^2.*

Proof. Flatness is automatic: ω is a closed 1-form on a 1-dimensional base. At a simple zero t_0 of Q_L: $Q_L(t) \approx Q_L'(t_0)(t - t_0)$, so $\omega \approx \frac{1}{2(t-t_0)}\,dt$, giving residue $\frac{1}{2}$. The monodromy exponentiation is immediate. For (iv): $\nabla^{\mathrm{sh}}\Phi = 0$ since $d\Phi = \omega\,\Phi$ by construction, and the evaluation at $t = 0$ gives $\Phi(0) = \sqrt{Q_L(0)}/(2\kappa) = 2\kappa/(2\kappa) = 1$. □

Remark 3.6. The shadow generating function $H(t) = 2\kappa\, t^2\, \Phi(t)$ is *not* a flat section of ∇^{sh}: the factor t^2 contributes $\nabla^{\mathrm{sh}}(H) \neq 0$. The flat section is $\Phi(t) = \sqrt{Q_L(t)}/(2\kappa)$; the shadow obstruction tower is the Taylor expansion of $t^2 \cdot \Phi(t)$, reflecting the degree offset (the tower starts at degree 2, not degree 0).

3.5. **The single-line dichotomy.**

Theorem 3.7 (Single-line dichotomy)**.** *Let L, S_r, α, Δ, Q_L be as above, and define the shadow depth $r_{\max}|_L := \max\{r : S_r \neq 0\}$ (with $r_{\max} = \infty$ if $S_r \neq 0$ for infinitely many r). Then $r_{\max}|_L \in \{2, 3, \infty\}$, classified by the shadow metric:*

(i) ***Class* G** *($\Delta = 0$, $\alpha = 0$). $Q_L = (2\kappa)^2$ is constant. $H(t) = 2\kappa\, t^2$. Only S_2 survives: $r_{\max} = 2$.*
(ii) ***Class* L** *($\Delta = 0$, $\alpha \neq 0$). $Q_L = (2\kappa + 3\alpha\, t)^2$ is a perfect square. $H(t) = t^2(2\kappa + 3\alpha\, t)$. S_2 and S_3 survive: $r_{\max} = 3$.*
(iii) ***Class* M** *($\Delta \neq 0$). Q_L is irreducible over $k(c)$; $\sqrt{Q_L}$ is irrational; $S_r \neq 0$ for all $r \geq 4$ generically: $r_{\max} = \infty$.*

For $r \geq 4$,

$$S_r = \Delta \cdot R_r, \qquad R_r \in \mathbb{Q}(\alpha, \kappa, S_4). \tag{3.8}$$

Proof. The classification follows from Theorem 3.1 and the Gaussian decomposition (Corollary 3.3).

Case $\Delta = 0$. Then $Q_L = (2\kappa+3\alpha t)^2$ is a perfect square, so $\sqrt{Q_L} = 2\kappa+3\alpha t$ is polynomial and $H(t) = t^2(2\kappa + 3\alpha t)$ terminates. If $\alpha = 0$: $H(t) = 2\kappa t^2$, giving $r_{\max} = 2$. If $\alpha \neq 0$: $H(t) = 2\kappa t^2 + 3\alpha t^3$, giving $r_{\max} = 3$.

Case $\Delta \neq 0$. The discriminant of Q_L as a polynomial in t is

$$(12\kappa\alpha)^2 - 4 \cdot 4\kappa^2 \cdot (9\alpha^2 + 16\kappa S_4) = -32\kappa^2\Delta \neq 0,$$

so Q_L is irreducible over $k(c)$. The Taylor expansion of $\sqrt{Q_L}$ around $t = 0$ has nonzero coefficients at all orders generically, hence $r_{\max} = \infty$.

Universal factorization. The binomial expansion of $\sqrt{Q_L}$ around the Gaussian envelope $(2\kappa + 3\alpha t)$ is

$$\sqrt{Q_L} = (2\kappa + 3\alpha t)\sqrt{1 + \frac{2\Delta\, t^2}{(2\kappa + 3\alpha t)^2}}.$$

Expanding $\sqrt{1+u}$ as $1 + \frac{1}{2}u - \frac{1}{8}u^2 + \cdots$ with $u = 2\Delta t^2/(2\kappa + 3\alpha t)^2$, every coefficient of t^n for $n \geq 2$ in $\sqrt{Q_L} - (2\kappa + 3\alpha t)$ carries a factor of Δ. Since $rS_r = [t^{r-2}]\sqrt{Q_L}$ for $r \geq 2$, and the Gaussian part $2\kappa + 3\alpha t$ contributes only at $r \leq 3$, the factorization $S_r = \Delta \cdot R_r$ for $r \geq 4$ follows. □

Remark 3.8 (The contact class). The partition **G**/**L**/**C**/**M** of chirally Koszul algebras includes a fourth class **C** (contact, $r_{\max} = 4$), realized by the $\beta\gamma$ system. This class escapes the single-line dichotomy by *stratum separation*: the quartic contact invariant lives on a charged stratum of $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}$ where $\kappa = 0$, and the shadow metric requires $\kappa \neq 0$. On the charged stratum,

the self-bracket maps to a zero-dimensional target by rank-one rigidity, so the quartic pump does not activate and the tower terminates at $r = 4$. The algebraic shadow depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ exhausts the four classes: **G** ($d_{\mathrm{alg}} = 0$), **L** ($d_{\mathrm{alg}} = 1$), **C** ($d_{\mathrm{alg}} = 2$), **M** ($d_{\mathrm{alg}} = \infty$).

3.6. **The shadow growth rate.** For class **M** algebras, the shadow obstruction tower is infinite but governed by a single analytic invariant: the growth rate of the coefficients S_r.

Definition 3.9 (Shadow growth rate)**.** Let L be a primary line of class **M** ($\Delta \neq 0$), and write $Q_L(t) = q_0 + q_1 t + q_2 t^2$ for the shadow metric. The *shadow growth rate* of L is

$$\rho_L \;:=\; \sqrt{\frac{q_2}{q_0}} \;=\; \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}. \tag{3.9}$$

By Vieta's theorem, the zeros t_0, $\bar{t}_0$ of Q_L satisfy $|t_0|^2 = q_0/q_2$, so $\rho_L = 1/|t_0|$: the reciprocal of the branch-point modulus. For classes **G** and **L**, set $\rho := 0$ by convention.

Theorem 3.10 (Shadow asymptotics)**.** *Let L be a primary line of class* **M** *with shadow growth rate ρ_L.*

(i) Asymptotic formula.

$$S_r \;\sim\; A(\mathcal{A})\, \rho_L^r\, r^{-5/2} \cos(r\theta + \varphi) \qquad (r \to \infty), \tag{3.10}$$

where $\theta := -\arg(t_0)$ is the oscillation phase, $A(\mathcal{A}) > 0$, and φ is computable from the branch-point residue of $\sqrt{Q_L}$. The exponent $-5/2$ arises from $S_r = [t^{r-2}]\sqrt{Q_L}/r$ combined with the $n^{-3/2}$ transfer law for square-root singularities.

(ii) Unit circle criterion. *$\rho_L = 1$ if and only if $q_0 = q_2$, i.e.,*

$$(2\kappa - 3\alpha)(2\kappa + 3\alpha) = 2\Delta. \tag{3.11}$$

Equivalently, the branch points of Q_L lie on the unit circle $|t| = 1$.

(iii) Convergence partition. *$\rho_L < 1$: the shadow obstruction tower converges absolutely. $\rho_L > 1$: the shadow obstruction tower diverges, but the algebraic function $\sqrt{Q_L(t)}$ extends analytically past the convergence radius. $\rho_L = 1$: marginal case (the critical surface).*

Proof. The function $\sqrt{Q_L(t)}$ has algebraic branch-cut singularities of order $\frac{1}{2}$ at each simple zero t_0 of Q_L. Since Q_L is quadratic with nonzero discriminant ($\Delta \neq 0$ implies $\mathrm{disc}(Q_L) = -32\kappa^2\Delta \neq 0$), the two zeros $t_0, \bar{t}_0$ are distinct with $|t_0| = |\bar{t}_0| = \sqrt{q_0/q_2} = 1/\rho_L$.

The transfer theorem for algebraic singularities [**?**, Theorem VI.1] gives $[t^n]\sqrt{Q_L} \sim C_0\, t_0^{-n}\, n^{-3/2}$ for each dominant singularity t_0 at distance $R = 1/\rho_L$ from the origin. Both branch points contribute (equal modulus), producing the cosine interference term from $2\,\mathrm{Re}(C_0\, t_0^{-(r-2)})$. Since $S_r = [t^{r-2}]\sqrt{Q_L}/r$ (Corollary 3.4), the extra factor $1/r$ gives the stated $r^{-5/2}$ exponent.

For (ii): $\rho_L = 1$ iff $q_2/q_0 = 1$. Writing $q_0 = 4\kappa^2$ and $q_2 = 9\alpha^2 + 2\Delta$, the condition $q_0 = q_2$ factors as $4\kappa^2 - 9\alpha^2 = 2\Delta$. Part (iii) is immediate from the definition of the convergence radius $R = 1/\rho_L$. □

3.7. **The depth decomposition.**

Theorem 3.11 (Depth decomposition)**.** *The shadow depth of a chirally Koszul algebra $\mathcal{A}$ decomposes as*

$$d(\mathcal{A}) \;=\; 1 + d_{\mathrm{arith}}(\mathcal{A}) + d_{\mathrm{alg}}(\mathcal{A}), \tag{3.12}$$

where:

(i) d_{arith} *counts the independent holomorphic Hecke eigenforms in the spectral decomposition of the completed shadow partition function $\widehat{Z}^c_{\mathcal{A}}$ on $\mathcal{M}_{1,1}$;*

(ii) $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ *is the* algebraic depth*, determined by the single-line dichotomy (Theorem 3.7) and stratum separation (Remark 3.8):*

$$d_{\mathrm{alg}} = \begin{cases} 0 & \textit{class } \mathbf{G}, \\ 1 & \textit{class } \mathbf{L}, \\ 2 & \textit{class } \mathbf{C}, \\ \infty & \textit{class } \mathbf{M}; \end{cases}$$

(iii) *depths $d \geq 5$ are purely arithmetic: they arise from d_{arith}, not from the A_∞ structure.*

Proof. The algebraic depth d_{alg} is determined by the structure of the cyclic deformation complex and the MC recursion: the single-line dichotomy (Theorem 3.7) gives $d_{\mathrm{alg}} \in \{0, 1, \infty\}$ on each primary line, and stratum separation (Remark 3.8) provides $d_{\mathrm{alg}} = 2$ for the contact class. Since $d_{\mathrm{alg}} \leq 2$ for finite cases ($\mathbf{G}/\mathbf{L}/\mathbf{C}$) and the base term is 1 (the curvature κ at degree 2), the maximum finite depth without arithmetic contributions is $1 + 0 + 2 = 3$; hence all depths $d \geq 5$ require $d_{\mathrm{arith}} \geq 2$.

The arithmetic depth counts independent holomorphic Hecke eigenforms by decomposing $\widehat{Z}^c_{\mathcal{A}}$ via the Roelcke–Selberg spectral decomposition on $\mathcal{M}_{1,1}$. Each holomorphic eigenform of weight k contributes a critical line at $\mathrm{Re}(s) = k/2$; Maass cusp-form projections contribute no new critical lines. □

Example 3.12 (Standard families)**.** The shadow metric data for the standard families:

Family	κ	α	S_4	Δ	Class	ρ
Heisenberg $\mathcal{H}_k$	k	0	0	0	$\mathbf{G}$	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{\dim \mathfrak{g}\,(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	$\mathbf{L}$	0
$\beta\gamma$ (weight λ)	$6\lambda^2{-}6\lambda{+}1$	0^*	0^*	0^*	$\mathbf{C}$	0
Virasoro Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	$\mathbf{M}$	$\frac{1}{c}\sqrt{\frac{180c+872}{5c+22}}$

* The table reports the weight-changing primary line, where $\kappa = S_3 = S_4 = 0$. The quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on this charged stratum; the shadow metric requires $\kappa \neq 0$ and does not apply. The class **C** designation ($r_{\max} = 4$) comes from rank-one rigidity, not the single-line dichotomy. The T-line (stress tensor) carries the Virasoro shadow data $(\kappa, \alpha, S_4) = (c/2, 2, 10/(c(5c{+}22)))$ with $c = 2(6\lambda^2{-}6\lambda{+}1)$ and is class **M**.

For the Virasoro algebra, the shadow metric evaluates to

$$Q_{\mathrm{Vir}}(t) = c^2 + 12c\,t + \frac{180c + 872}{5c + 22}\,t^2,$$

with Gaussian decomposition $Q_{\mathrm{Vir}} = (c + 6t)^2 + \frac{80}{5c+22}\,t^2$. The critical central charge c^* where $\rho = 1$ is the unique positive real root of

$$5c^3 + 22c^2 - 180c - 872 = 0,$$

approximately $c^* \approx 6.12$. For $c > c^*$ the shadow tower converges; for $c < c^*$ it diverges. At the Koszul self-dual point $c = 13$: $\rho(\mathrm{Vir}_{13}) \approx 0.467$.

Remark 3.13 (Spectral curve). The algebraic relation $H^2 = t^4 Q_L(t)$ defines, for each primary line L, a spectral curve $\Sigma_L := \{H^2 = t^4 Q_L(t)\} \subset \mathbb{A}^2_{t,H}$. Since Q_L is quadratic in t, Σ_L has arithmetic genus 0; the projection $(t, H) \mapsto t$ is a double cover ramified at the zeros of Q_L. The degeneration loci in parameter space are controlled by the critical discriminant: Σ_L acquires a node precisely when $\Delta = 0$. The shadow growth rate ρ_L is the reciprocal of the distance from $t = 0$ to the nearest ramification point.

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